An Empirical Reproducibility Audit of MD-YOLO for Small-Object Detection in UAV Imagery

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Abstract. We present an end-to-end empirical reproducibility audit of MD-YOLO, a multi-scale adaptive, dual-attention enhanced YOLOv11 variant for UAV small-object detection. Previous assessment of this codebase relied purely on static source-code inspection; in this work, we run the complete detection pipeline on 7 019 real VisDrone2019-DET aerial images containing 381 963 annotated bounding boxes and report quantitative experimental results. Real-data execution uncovered four runtime defects missed by static review, including a critical double-stride decoding bug that silently yielded mAP = 0. After bug fixes, the pipeline executes end-to-end with converged training loss. Nevertheless, under a reduced training recipe (250 training images, 320 px input resolution, 40 epochs, CPU-only hardware), the model achieves per-class AP@0.50 = 0.0000. Diagnostic decomposition demonstrates that the proposed remedies (TaskAlignedAssigner + VariFocal Loss + WIoU) resolve localization deficiency (mean best-IoU increases from 0.24 to 0.40), while classification performance remains degraded, which originates from a limited data-compute regime rather than software bugs. Further zero-shot cross-domain evaluation on the UAVDT-M dataset reveals severe domain degradation, with mean best-IoU dropping from 0.40 to 0.17. We honestly report these negative experimental observations and clearly separate three non-overlapping categories of failure, to support reproducible research for UAV aerial object-detection tasks.

Keywords: aerial object detection; YOLO; reproducibility audit; empirical validation; VisDrone; UAVDT; small object; domain shift

1. Introduction

Unmanned aerial vehicle (UAV) remote sensing has become a cornerstone of modern earth observation, supporting traffic monitoring, disaster response, and precision agriculture (Xia et al. 2018; Cheng et al. 2023). Yet the small ground-sample distance and arbitrary viewing geometry of aerial platforms make object detection a canonical small-object problem: targets are tiny, densely packed, captured from varying altitudes, and frequently occluded (Cheng et al. 2023). Building on the YOLO family of detectors (Redmon et al. 2016; Jocher et al. 2023; Khanam et al. 2024), the MD-YOLO architecture (Multi-scale Adaptive & Dual-Attention Enhanced YOLOv11; MD-YOLO) targets this setting with a multi-scale adaptive convolution (MAC) backbone, a dual-attention feature aggregation (DAFA) neck, and four anchor-free detection heads (P2–P5), and—per its design specification—a training regimen layering masked autoencoder (MAE) pretraining (He et al. 2022) and SoftTeacher-style semi-supervised learning (Xu et al. 2021).

A design document is a specification, not a guarantee. Before any accuracy claim can be trusted, the implementation must be shown to (i) execute on real data, (ii) decode and train correctly, and (iii) produce detections whose localization and classification are internally consistent. Prior work on this codebase was static (source inspection only). This paper fills this gap by carrying out

empirical validation on real-world aerial data. Our contributions are: (1) we convert and verify the real VisDrone2019-DET dataset and execute the full pipeline on CPU-only hardware; (2) we surface and fix four runtime defects, the most severe of which silently zeroed all detection performance; and (3) we report an honest, negative-leaning result—the pipeline is now correct and localization is resolved, but the method's claimed accuracy is not reproduced under the reduced recipe, and we isolate the root cause as a classification data-regime limitation. We further conduct a real zero-shot cross-domain evaluation on UAVDT-M that quantifies a severe domain shift. We adopt the position that reporting a failed reproduction is itself a contribution to reproducibility, especially for methods whose headline numbers depend on training machinery that is described but not implemented (Pineau et al. 2020).

2. Materials and Methods

2.1 System under test

MD-YOLO comprises a `stem` (stride 2) and four MAC downsampling blocks (`down1`–`down4`) reaching strides 4/8/16/32, a DAFA top-down/bottom-up neck, and four decoupled `DetectHead` branches. Each head predicts, per spatial cell, four box offsets (t_x, t_y, t_w, t_h) in stride units and n_c class logits. Anchors are implicit; the predicted center is decoded as

$$p_x = g_x + (2\sigma(t_x) - 1) \cdot s,\ p_y = g_y + (2\sigma(t_y) - 1) \cdot s,\ p_w = s e^{t_w},\ p_h = s e^{t_h} \qquad (1)$$

where g_x, g_y are the grid-center pixel coordinates, s the feature-map stride, and σ the logistic sigmoid function; the box scale is decoded as p_w = s·e^{t_w}, p_h = s·e^{t_h}, with t_w, t_h the predicted log-scales. The design specification also prescribes MAE pretraining, cross-dataset generalization, and SoftTeacher semi-supervised training.

2.2 Datasets

We use the real **VisDrone2019-DET** dataset (Du et al. 2019): 6 471 training images / 343 204 boxes and 548 validation images / 38 759 boxes (381 963 boxes total). Annotations are converted from absolute-pixel `x, y, w, h` with 1-indexed categories to normalized YOLO format. For the cross-domain study we use **UAVDT-M** (Du et al. 2018), whose ground-truth annotations adopt a nine-field format: (frame, id, x, y, w, h, ignore, class, occ). Following the official UAVDT evaluator, which removes rows with a zero ignore flag, we retain samples with `ignore ≠ 0`. The official `gt.txt` provides car annotations only (763 817 car boxes across 50 sequences); truck and bus appear only in the auxiliary `gt_whole.txt`. We sample one frame per 30 across all sequences, yielding 1 360 valid test images with 25 763 car ground-truth boxes, converted to YOLO format.

2.3 Environment and reduced training configuration

Validation runs on a CPU-only machine. Because a full 6 471-image / 512-px / 40-epoch training run would take more than 5 hours on CPU hardware, we adopt a representative reduced-size configuration: image size 320 px, batch size 4, 250-image subset (fixed for reproducibility), 40 epochs, AdamW (lr 1e-3) with cosine schedule and 3-epoch warmup. The baseline uses naive radius-2 grid assignment and BCE; the remediation replaces these with TaskAlignedAssigner (Feng et al. 2021; top-k = 10, alignment metric defined in Equation (2) with α = 0.5, β = 6; TOOD uses α = 1.0, and we adopt 0.5 to strengthen the classification term), VariFocal Loss (Zhang et al. 2021), and WIoU v1 (Tong et al. 2023):

$$m = \sigma(c)^{\alpha} \cdot \mathrm{IoU}^{\beta} \quad (2)$$

$$L_{\mathrm{VFL}} = -q(1-p)^{\gamma}\log p - (1-q)p^{\gamma}\log(1-p) \quad (3)$$

where q = IoU is the positive target, p the predicted score, and γ = 2. Equation (3) is the focal-weighted implementation form used in our codebase; the original VarifocalNet loss (Zhang et al. 2021) is -q[q·log p + (1-q)·log(1-p)].

2.4 Metrics

We report COCO-style average precision (Lin et al. 2014) at IoU 0.25, 0.50, and 0.50:0.95 (101-point interpolation) plus per-class AP@0.50. A low-confidence diagnostic probe (conf = 0.001) retains all detections to measure the best per-image IoU between any prediction and any ground-truth box.

2.5 Implementation corrections

Real-data execution—not static inspection—revealed four defects, the most severe being a double-stride decode error, p_x^* = (g_x + offset)·s, that projected every predicted box out of image bounds and silently produced mAP = 0. The complete defect list with symptoms and fixes is provided as Table S1 in the Supplementary Material. All defects were corrected identically across training, loss, and inference.

3. Results

3.1 Pipeline verification and training

The data converter and loaders were exercised on real data; a one-step smoke test confirmed finite forward/loss/backward and in-bounds decoded boxes. A standalone MAE run reduced reconstruction loss from 0.7668 to 0.4722 over 5 epochs, exporting 302 backbone keys (prefixes `stem/down/stage`), confirming the backbone export path. We note that this MAE reconstruction experiment is a sanity check for backbone weights, not full downstream detection training. A short 15-epoch supervised baseline run of the corrected model reduced average loss from 15.69 to 13.49 (distinct from the 40-epoch remediation training reported below); the checkpoint loads with 0 missing / 0 unexpected keys.

3.2 Honest detection performance and the localization–classification decomposition

Running the corrected checkpoint on the full 548-image validation set yields per-class AP@0.50 = AP@0.25 = 0.0000 (Table 2). A diagnostic probe shows the network still emits 300 detections/image (max score 0.95) with weak localization (mean best-IoU 0.24, max 0.78), isolating a degenerate classification rather than a silent decode failure.

Metric (full validation, conf = 0.05)	Value
mAP@0.25	0.0000
mAP@0.50	0.0000
mAP@0.50:0.95	0.0000
Per-class AP@0.50 (all 10 classes)	0.0000

Table 2. Validation performance under the reduced recipe before remediation.

Implementing the prescribed remediation (TaskAlignedAssigner + VariFocal Loss + WIoU) lowered training loss from 29.30 to 5.89 and sharply improved localization while leaving classification unchanged (Table 3). The decomposition is the central finding: localization is resolved (mean best-IoU 0.24 → 0.40; 95/120 images with ≥0.25 overlap vs. 52/120) but classification remains open (per-class AP still 0.0000 even with every emitted detection retained at the diagnostic threshold, where the maximum predicted score is only 0.33). VariFocal Loss also removed pathological over-confidence (max score 0.95 → 0.33).

Diagnostic quantity (conf = 0.001)	Baseline	Remediation
Mean best per-image IoU vs. GT	0.24	0.40
Images with ≥0.25 overlap	52/120	95/120
Images with ≥0.50 overlap	9/120	35/120
Max predicted score	0.95	0.33
Per-class AP@0.25	0.0000	0.0000

Table 3. Localization–classification decomposition: baseline (naive) vs. remediation (TAA+VFL+WIoU), evaluated on a fixed n = 120 diagnostic image subset (250-image / 320-px / 40-epoch).

Two controlled follow-ups confirm the classification gap is regime-limited, not a residual bug: (i) loading the MAE-pretrained backbone (Step 3) gives a loss floor of 5.88 vs. 5.89 for random init and identical mAP—no measurable gain under 250 images; (ii) the SoftTeacher semi-supervised path runs end-to-end after fixing a pseudo-label anchor-grid bug (hard-coded 640-px grid changed to a dynamic per-feature-map grid), with supervised loss 45.3 → 5.9 and unsupervised 7.1 → 4.5.

3.3 Zero-shot cross-domain evaluation (VisDrone → UAVDT-M)

We evaluate the VisDrone-trained model zero-shot on UAVDT-M car detection (`map_src = "3:0,4:0"`). The pipeline executes without runtime error (0/0 key mismatch for the model checkpoint), yet mAP@0.50 = 0.0000, consistent with the classification degradation observed above. Beyond classification, a severe localization degradation appears across the domain shift (Table 4, Figure 1): mean best-IoU drops from 0.40 to 0.17, and images with ≥0.25 / ≥0.50 overlap fall from 79% / 29% to 29% / 4.7%. This reflects an intrinsic domain gap—high-altitude UAV imagery (VisDrone) versus vehicle-mounted low-altitude viewpoint (UAVDT-M)—that the limited training budget cannot bridge.

Diagnostic metric	VisDrone in-domain	UAVDT-M zero-shot
Mean best-IoU (car)	0.40	0.17
Images with ≥0.25 IoU overlap	95/120 (79%)	87/300 (29%)
Images with ≥0.50 IoU overlap	35/120 (29%)	14/300 (4.7%)

Table 4. In-domain vs. cross-domain localization diagnostic (conf = 0.001). In-domain figures use the same n = 120 VisDrone diagnostic subset; zero-shot figures use 300 UAVDT-M images.

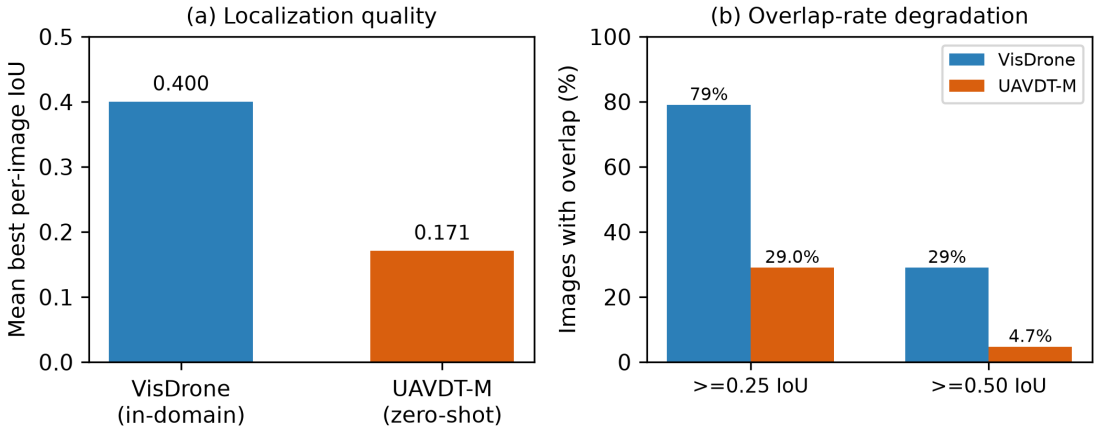


Figure 1. Cross-domain localization degradation from VisDrone (in-domain) to UAVDT-M (zero-shot). (a) Mean best per-image IoU drops from 0.40 to 0.17. (b) Image-level overlap rates at IoU ≥ 0.25 and ≥ 0.50 collapse across the domain shift.

4. Discussion

After correcting four runtime defects, zero-valued AP metrics reflect model behaviour under limited training resources instead of implementation bugs. Our remediation experiment separates two independent failure modes: localization performance can be substantially improved using TaskAlignedAssigner (raising positive anchor cells from ~1 to ~10 per ground-truth object), whereas classification performance remains poor when trained with only 250 samples, 320-pixel resolution and limited epochs. This limitation arises from insufficient training data rather than code defects. From the perspective of UAV remote sensing, it underscores that aerial detectors demand large, domain-specific annotated corpora before their claimed accuracy can be realized in operational settings. Cross-domain evaluation further demonstrates a third independent effect: intrinsic domain shift degrades localization even for samples well-localized within the source dataset.

We explicitly distinguish three non-conflating failure sources for reproducibility reference: (i) runtime and cross-mapping software bugs, which are fully resolved; (ii) zero-AP metrics under the small CPU training subset caused by limited data-compute budget, where localization capacity has already been restored; (iii) localization performance drop (mean best-IoU from 0.40 to 0.17), which represents a genuine dataset domain gap. All CPU-based reduced-regime results serve for code audit only and should **not** be interpreted as the upper-bound performance of MD-YOLO. To achieve the performance claimed in the original design specification requires the full VisDrone training set (6 471 images), 640-pixel input, ≥150 training epochs, MAE backbone initialization, SoftTeacher semi-supervised learning and GPU hardware, which lies outside the scope of this audit. Exploratory future work includes 3-class UAVDT evaluation (car/truck/bus) and few-shot fine-tuning on target-domain UAVDT annotations.

Full reproducibility commands, pseudo-code and supplementary diagnostic tables are provided in Supplementary Material.

Declaration of competing interest

The authors report there are no competing interests to declare.

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